A semi-supervised learning approach for AI content detection on Instagram

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Abstract

With generative artificial intelligence (AI) becoming increasingly realistic, detecting AI-generated content on social media is becoming increasingly difficult, blurring the line between real and fake. In this study, we use a small dataset of real and AI-generated images from Instagram to investigate whether metadata - technical information and platform-specific data belonging to images - could help classify content as either real or AI-generated. Due to the limited dataset size, we use a semi-supervised learning (SSL) approach to pseudo-label images, thereby increasing the amount of training data. We integrate this approach into a pre-trained Convolutional Neural Network (CNN) and finetune this CNN using iterative loops. We find that SSL in combination with finetuning increases the performance of the CNN model, which uses visual input to classify content. Next, we compare this finetuned CNN with SSL to metadata-augmented variants. Specifically, we investigate the contributions of EXIF metadata, social media metadata, and both types of metadata combined. We find that the CNN model that uses both types of metadata as input achieves the highest performance (accuracy = 0.878, precision = 0.769, recall = 1.000, F1 score = 0.867). These results show that metadata can indeed improve a CNN model's classification performance 'in the wild'. Once more studies are conducted on larger datasets and performance is increased even more, we believe this pipeline could be employed by social media platforms to help classify social media content.

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1 Introduction

As generative Artificial Intelligence (AI) technologies continue to improve, distinguishing real visual content from AI-generated material has become increasingly difficult. Recent studies show that even human observers struggle with this task, correctly identifying AI-generated images only slightly above chance level [1, 2]. This increasing realism of AI complicates how online content is perceived and trusted, especially in digital environments where misleading information can easily be mistaken for true information [3]. Consequently, being able to reliably detect AI-generated images has become increasingly important. Detection is a necessary first step for enabling practical responses, such as content labeling, authentication mechanisms, and transparency measures that aim to reduce the societal risks of synthetic media [3, 4].

At the same time, detecting AI-generated images on social media is far from straightforward. Images are often compressed, resized, edited, or stripped of technical metadata during the upload process, which removes many potentially useful detection cues [3]. While numerous detection approaches have been proposed, many focus on video content, where models can exploit temporal inconsistencies across consecutive frames. In contrast, image-based detection relies solely on spatial and visual evidence from a single image, making the task inherently more difficult. As a result, methods that perform well in controlled settings often degrade when applied to real-world social media data [3].

As discussed by Verdoliva et al. [3], social media images are not limited to visual information alone, but are accompanied by a wide range of metadata that may provide useful signals for AI-generated image detection. This metadata includes both platform-level information and technical attributes embedded within image files, such as EXIF metadata. However, there remains relatively little empirical work that systematically investigates how visual features and non-visual cues can be combined to improve AI-generated image detection in uncontrolled social media environments. Motivated by this gap, our study addresses how such metadata could be combined with Convolutional Neural Networks (CNNs), which have proven effective in learning visual patterns from pixel-level data. Focusing solely on images taken from Instagram, our study will address the following research question:

RQ1: How does the inclusion of metadata affect the performance of CNNs in detecting AI-generated images on Instagram?

We hypothesize that incorporating metadata alongside visual features will improve the detection of AI-generated images compared to using visual information alone. We will investigate how both social media metadata and EXIF metadata contribute to changes in detection performance and whether a synergistic effect occurs when both types of metadata are incorporated.

A big challenge that makes generative AI content detection "in the wild" even more difficult is the lack of reliable ground-truth labels for images on social media. On Instagram, posts rarely come with verified labels, and even manual annotation can be time-consuming and uncertain. This limitation also affects our dataset, where only a small subset can be labeled with high confidence,

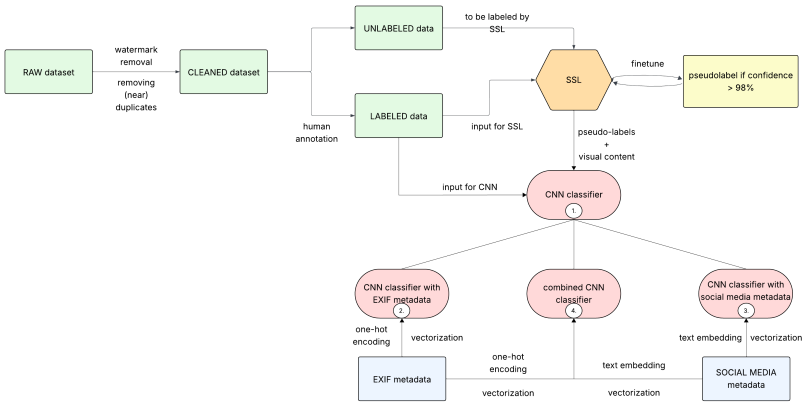


Figure 1: Experimental pipeline of this study. This study will compare the classification performance of AI content detection using 1. only a (finetuned) CNN, 2. a CNN with EXIF metadata, 3. a CNN with social media metadata, and 4. a CNN combining both EXIF and social media metadata.

which leads us to our second research question:

RQ2: How can a semi-supervised learning approach be employed to more reliably label unlabeled instances for use in training a final classifier for AI image detection in the wild?

In this study, we adopt a semi-supervised learning (SSL) approach, which reduces the amount of manual labeling required. We aim to study how unlabeled images can be incorporated more reliably through confidence-based pseudo-labeling when training a final CNN classifier. With SSL, the model iteratively looks at unlabeled images and assigns pseudo-labels to them when it is confident in its predictions. These model-assigned labels are then added to the training data and used in subsequent training rounds. This process allows the model to gradually increase the amount of labeled data and improve its performance over time. We will investigate the performance change by comparing a CNN model without an SSL approach to one with an SSL approach.

In Figure 1, we show the experimental pipeline of our study that - based on a few hand-labeled images - employs semi-supervised learning to expand the amount of labeled training instances. Then, using this expanded dataset, it combines CNN-based visual feature extraction with incorporated metadata analysis. We evaluate the performance of a CNN-based classifier under different experimental settings: 1. using only visual features, 2. combining visual features with EXIF metadata, 3. combining visual features with social media metadata, and 4. combining visual features with both metadata types. Based on this we will see which metadata is most effective in improving AI detection and hopefully get further insight into which future steps can be taken to improve our pipeline.

2 Research Context

2.1 CNNs for AI-Generated Content Detection

Previous work on AI-generated image detection has explored a range of deep learning models, including Vision Transformers and CNN-based detectors. Among these, CNN-based approaches remain widely used as strong and data-efficient baselines, particularly in settings with limited labeled data and high visual variability, such as social media images [13, 14]. CNNs learn hierarchical visual features through convolutional filters, allowing them to capture generation-related patterns such as unusual textures, inconsistent colors, and certain frequency signals. [16, 17, 18, 19]. Unfortunately, CNN performance can drop on real social media images due to compression, resizing, and other platform-specific processing [20, 21]. Similar limitations are likely to affect other pixel-based deep learning models. We therefore hypothesize that platform-level metadata which is often preserved despite image recompression and editing can provide complementary information that improves CNN-based detection performance in real-world social media settings.

2.2 Metadata as a Signal for AI-Generated Content

In social media environments, images are not standalone visual objects but are shared with technical information and platform-specific metadata. While most existing AI-generated image detection approaches rely mainly on visual features extracted by CNN-based detectors, such methods may overlook non-visual cues that can provide important information about how an image was created and shared.

In this project, we therefore incorporate metadata alongside visual features. This choice is motivated by the observation that generative pipelines and social media platforms often leave traces that are not directly visible at the pixel level, especially in real-world social media settings where images are frequently compressed or resized.

We consider two main types of metadata: EXIF metadata and social media metadata. EXIF metadata consists of technical attributes embedded in image files, such as camera model, software used for image creation or editing, image resolution, color space, and timestamps [7]. Real photographs often contain EXIF information that reflects physical camera hardware and capture conditions, whereas AI-generated images may show missing, inconsistent, or implausible EXIF fields, particularly with respect to camera identifiers and software tags [6]. However, on platforms such as Instagram, EXIF metadata is often partially stripped or altered due to compression and re-encoding, making it unreliable on its own but still potentially informative when combined with other signals [11, 12].

By contrast, social media metadata refers to platform-level information associated with each post rather than the image file itself. This includes attributes such as caption length, number of hashtags, hashtag categories, posting time,

engagement metrics, and contextual descriptors inferred from the post [9]. AI-generated content on Instagram is often associated with distinctive posting patterns, such as excessive hashtag usage, explicit AI-related tags (e.g., *#ai*, *#aiart*, *#genai*), or atypical engagement behavior [12]. Unlike EXIF metadata, these platform-level features are not affected by image compression and therefore provide a complementary signal for AI-generated image detection in the wild.

This directly motivates our first research question (RQ1), which examines whether combining EXIF and social media metadata with visual features improves the detection of AI-generated images on Instagram.

2.3 Legal Perspective on AI-Generated Images

In the context of synthetic media, a broader legal and regulatory challenge concerns determining when transparency measures, such as mandatory labeling or disclosure, should be applied to AI-generated content. This question is not straightforward, as not all AI-generated images pose the same level of risk or potential for harm. In current regulatory discussions, including the EU AI Act, particular attention is given to so-called deepfakes, as these forms of synthetic media are more likely to mislead viewers or create false impressions[24].

A related and more specific challenge, therefore, lies in defining what qualifies as a deepfake. An AI-generated image or video is generally considered a deepfake when it realistically mimics or replaces a real person (or, in some cases, an existing place or event) in a way that could reasonably be mistaken for reality [3, 26]. This also includes manipulated videos that imitate real voices, facial expressions, or behaviors [26]. In contrast, AI-generated content that does not involve a real person, such as synthetic faces or abstract imagery, is generally classified as AI-generated rather than as a deepfake.

For these reasons, we adopt a simple and practical labeling scheme that reflects current legal considerations under the EU AI Act and aligns with established practices in deepfake detection benchmarks, as detailed in Section 3. While the EU AI Act mainly targets transparency measures at the content creation stage, the techniques studied in this work could support such objectives both at creation time and at the platform level, for example, by enabling content labeling in social media contexts. This perspective is also consistent with broader transparency goals discussed in the Digital Services Act. [24, 25]

3 Data Pre-processing

In this section, we describe the dataset and explain how the images were collected, labeled, and processed to serve as input for the proposed semi-supervised learning framework and CNN model. Careful dataset construction was necessary to reduce overfitting and shortcut learning. Without manual cleaning, redundancy removal, and agreement-aware labeling, the model could learn spurious correlations, such as associating watermarks, repeated templates, or hashtag-specific artifacts with the AI-generated class. By prioritizing high-confidence la-

bels and maintaining consistency across labeled and unlabeled data, the dataset provides a robust foundation for the proposed semi-supervised learning framework.

3.1 Data Sources

The dataset used in this study was collected from publicly accessible Instagram posts. It was designed to reflect a realistic social media environment in which AI-generated and human-generated images coexist, while maintaining sufficient label reliability for supervised and semi-supervised learning.

Human-generated (real) images were obtained using location-based searches. Specifically, 180 real-labeled images were collected from posts tagged with the locations *Amsterdam*, *Maastricht*, *Berlin*, *Copenhagen*, and *St. Gallen*. Each image was evaluated independently by two team members before being placed in the *real* folder.

Second, AI-generated images were collected using hashtag-based searches commonly associated with generative AI content. Images were gathered from different subfolders of the *Hashtag* directory, corresponding to hashtags such as *ai*, *aiart*, *aicat*, *aigenerated*, *aitrends*, *aivideo*, *artificialintelligence*, *genai*, and *trumpai*. Creators frequently use these hashtags to indicate AI-generated or AI-assisted content. From these sources, two annotators verified 57 images and included them in the AI-labeled subset. Visible watermarks were cropped out prior to training.

In addition to the labeled data, a large set of unlabeled images was retained from the remaining hashtag-based content and the Will Buyers bucket, resulting in a total of 1228 unlabeled images.

3.2 Labeling Categories

The images were assigned to one of three labeling categories:

- **Real**: images assumed to originate from natural image capture processes
- **AI-generated**: images produced using generative AI models
- **Uncertain**: images whose origin could not be determined with high confidence

Only images labeled as *Real* or *AI-generated* were included in the labeled dataset. Images categorized as *Uncertain* were excluded from the labeled set and treated as unlabeled data to avoid introducing label noise into the semi-supervised training. This labeling strategy is consistent with established practices in widely-used deepfake detection benchmarks, where datasets distinguish between real and synthetic media and exclude ambiguous samples from confident labeling [23].

Images labeled as *Uncertain* were intentionally retained rather than discarded. The motivation for this decision was to better understand how the

proposed model handles ambiguous cases that are difficult even for human annotators to classify. By keeping these images as part of the unlabeled dataset, we are able to observe how the algorithm evaluates and assigns confidence to uncertain content during the semi-supervised learning process.

3.3 Manual Annotation and Annotator Collaboration

All labeled images were manually annotated, with each image being independently reviewed by at least two team members familiar with common characteristics of AI-generated content. Only images for which both annotators initially agreed were included in the labeled dataset. In cases of disagreement or uncertainty, images were jointly reviewed to assess whether a confident label could be assigned. If no consensus was reached, the image was labeled as *Uncertain* and excluded from the supervised training set. Similar multi-annotator and consensus-based annotation strategies have been adopted in prior work to ensure label reliability in AI-generated content detection [23].

Inter-annotator agreement was quantified on the *initial independent annotations*, before any discussion, using Cohen's kappa [22]. Cohen's kappa is defined as

$$\kappa = \frac{p_o - p_e}{1 - p_e},$$

where p_o denotes the observed agreement between annotators and p_e the agreement expected by chance. Based on our annotation statistics (227 labeled images with 4 initial disagreements), this yields $\kappa \approx 0.953$, indicating a near-perfect level of agreement. This result suggests that the independently assigned labels are highly reliable.

Cohen's kappa was used as an indicator of annotation consistency rather than as the sole criterion for label quality. The distinction between AI-generated and human-generated images is inherently ambiguous and continues to evolve as generative models improve. Consequently, some disagreements reflected genuine uncertainty rather than annotation errors. In such cases, consensus-based review was applied, and images for which no confident agreement could be achieved were retained as unlabeled data for use in the semi-supervised learning framework.

3.4 Watermark Removal and Data Cleaning

To prevent the CNN from relying on trivial visual cues, all visible watermarks and logos were removed from both labeled and unlabeled images. Watermarks constitute a strong confounding factor in AI-generated content detection, as many generative tools embed characteristic signatures that may lead models to exploit shortcut learning rather than meaningful visual patterns [5].

Watermark removal was performed manually on an image-by-image basis. When watermarks were located near image borders, images were cropped to eliminate the artifact. When watermarks appeared in central regions, they were masked or inpainted in order to preserve the surrounding visual content. This

cleaning procedure was applied consistently across both labeled and unlabeled images to avoid introducing distributional differences between data splits. Examples of the watermark removal process are provided in Appendix A.1.

3.5 Challenges in Dataset Construction

One major challenge during dataset construction was the high visual similarity between many AI-generated images. In many cases, multiple images were generated from the same source with only small variations, such as changes in pose, background, or viewpoint. As a result, a significant portion of the dataset consisted of near-duplicate images.

This posed difficulties when splitting the data into labeled, unlabeled, training, and test sets. If visually similar images appear in both training and test data, the model may achieve artificially high performance by memorizing specific patterns rather than learning general characteristics of AI-generated content. At the same time, aggressively removing similar images reduces dataset size, which is particularly problematic for the limited labeled data. We tried to balance this by handpicking images in the test set that did not have near duplicates in the training set and keeping a maximum of five near-duplicates in the training data.

4 Methodology

4.1 Semi-Supervised Learning

This section describes the motivation for using semi-supervised learning, the dataset splitting strategy, the base CNN model, and the design choices made to ensure reliable pseudo-labeling under class imbalance.

4.1.1 Overview of Semi-Supervised Learning

In order to make use of our dataset containing mostly unlabeled images, we use semi-supervised learning, as explained in Algorithm 1. This approach uses fine-tuning and pseudo-labeling in order to incorporate unlabeled data into the training process. At each iteration, the model is finetuned on labeled data. It then attempts to predict the labels of the unlabeled instances, and in case it can do so with a confidence exceeding a threshold of 0.98, its prediction is then used as a pseudo-label at the next iteration of the finetuning process. We run three iterations of the semi-supervised learning before the last round of finetuning on the final dataset.

4.1.2 Dataset Splitting and Evaluation Protocol

Before we can begin with the training, it is important to make sure that we correctly split our dataset into a training, validation, and test set. The test set is handpicked to ensure all metadata for it is available (i.e., both EXIF and

social media metadata), so that we are able to test all models on the same test set. The remaining labeled items are split into training (80%) and validation (20%) using stratified sampling to ensure that precision and recall metrics are not skewed by a different class distribution. At this stage, we also establish the positive class. Since the ultimate goal is to detect AI samples, we decided to assess AI as the positive class. Furthermore, since the AI class is the minority class, this will make our precision and recall results much more significant, as each misclassified instance will be strongly reflected in the metrics.

4.1.3 CNN Model Variants

To implement semi-supervised learning, we first need a base model for which the semi-supervised learning loop would be run. Since the amount of labeled data we have is insufficient to train a CNN model from scratch, we resort to using a pre-trained model. For this, we choose the "AI vs Human Image Detector" model provided by Ateeqq on Hugging Face [27]. Since we use data that is newer than the release of the model above, the model is unlikely to overfit to our data and the new data should improve the model's ability to generalize. We test this model out-of-the-box to set a baseline for how this model performs on social media images. Next, we compare this out-of-the-box model with 1. the same CNN model finetuned to our data but without an SSL loop, and 2. our finetuned CNN model with an incorporated SSL loop. The performance of these models can be found in Table 1 in the Results section.

4.1.4 Pseudo-Labeling Strategy and Training Loop

In order for the pseudo-labels to be at least to some extent reliable, we need an accuracy and precision that would let us believe that the pseudo-labeled data is indeed useful, and not just noise. Therefore, at each iteration of the semi-supervised learning loop, we finetune the CNN model on the currently available data, as described in Algorithm 2. The fine-tuning step enables us to improve the performance of the CNN, yielding higher-quality labels. We compute training and validation set metrics (accuracy, precision, recall, F1 score) in order to ensure that the model does not overfit. We also use the validation set to assess which confidence threshold for assigning pseudo labels results in the best model performance. We run three iterations of the semi-supervised learning loop, followed up by a final fine-tuning round on the resulting dataset. Once this completes, we then also compute test set metrics to assess the impact of finetuning.

4.1.5 Weighting of Instances

Furthermore, we also need to address the class imbalance that is present in the dataset. Therefore, we assign a class weighting to ensure that the model is not skewed towards classifying instances as not AI (i.e., real). The approach we suggest considers the expected loss from misclassifying an instance. For AI

versus not AI, we wish to weight the AI class in such a way as to ensure that misclassifying an instance of the AI class as not AI results in the same loss as misclassifying not AI as AI. Therefore, we multiply the loss of each misclassified AI instance by a scaling factor larger than one in order to account for the class imbalance. To calculate the scaling factor, we first calculate the class balance of the training set. Then, we assign the weight of the not AI class as 1, and the weight of the AI class as $\frac{n_{notai}}{n_{ai}}$. In practice, we then round this number to avoid numerical stability and rounding issues. We also ran the loop with other values for this hyperparameter, yet we found the approach above to have the best performance on the validation set.

We also want to account for the fact that pseudo-labels are not as reliable and scrutinized as ground truth labels. Even though the model needs to have a very high confidence (above 0.98) for a pseudo-label to be accepted, these labels may still be erroneous. Hence, we wish to assign greater weight to human-labeled training instances as opposed to pseudo-labeled training instances. Therefore, our suggested approach is to weight the pseudo-labels as 0.3 times as important as true labels, to ensure that wrong pseudo-labels do not dominate the model. This value was chosen as it led to the best performance on the validation set out of a range of values between 0 and 1 that we tested.

4.2 Metadata Extraction

Metadata extraction was performed using separate pipelines for EXIF metadata and social media metadata. This separation was necessary due to the different sources and formats. These fields capture file-level properties, image encoding characteristics, and compression-related artifacts, rather than camera-specific parameters that were found to be removed by Instagram's re-encoding process.

For the EXIF metadata, we retained a subset of fields that were consistently extractable across the dataset and potentially informative for distinguishing between human-generated and AI-generated images, using the ExifTool extraction framework. Specifically, the extracted EXIF features include file identifiers and storage information (e.g., *filesize*), which can reflect compression behavior. Image-level properties such as *Megapixels*, *ImageSize*, *ColorComponents*, and *YCbCrSubSampling* were extracted to capture structural and color encoding characteristics that may differ between natural photographs and synthetically generated images.

In addition, encoding-related metadata fields such as *EncodingProcess* and resolution parameters (*XResolution*, *YResolution*) were included to reflect how images were compressed and formatted prior to upload. These properties can encode subtle differences introduced by generative tools, post-processing software, or platform-specific image pipelines.

Finally, a discrete quantization table fingerprint (*dqt_fingerprint*) was extracted to capture JPEG compression artifacts at a low level. Since many AI image generators and editing tools rely on characteristic compression pipelines, this feature provides a compact representation of image encoding behavior that is not directly observable at the pixel level.

Social media metadata was obtained from an existing dataset provided alongside a subset of the collected Instagram images. Unlike EXIF metadata, which was extracted directly from image files, social media metadata was available only for a limited portion (530 images) of the dataset. The available social media metadata included both textual and numerical attributes describing the context of each Instagram post, such as caption-based features, hashtag-related statistics, and basic posting information, such as like count. Since this metadata originated from an external source, its structure and availability were fixed and could not be recovered for the remaining images.

All the metadata were collected and consolidated into a unified tabular representation, ensuring a one-to-one correspondence between each image and its associated metadata through filename-based alignment.

4.3 Metadata-Enhanced Models

To enhance the classification model with contextual signals beyond pixel-level information, image metadata was incorporated into the learning pipeline. Three metadata-aware multimodal models were developed (see Figure 1). The first model integrates intrinsic image metadata derived from EXIF information, while the second incorporates external social media metadata reflecting the context in which images are shared. The third model combines both of these into a single model.

Since metadata information is available only for a subset of the dataset, pseudo-labels were leveraged to increase the number of training samples in the metadata-enhanced models. In particular, pseudo-labeled images for which metadata were present were included, allowing the multimodal models to benefit from a larger effective training set.

4.3.1 CNN Architecture with EXIF Metadata Integration

Structured EXIF metadata were incorporated into the CNN architecture. Both numerical and categorical attributes were included. Numerical features (e.g., file size, megapixels, and resolution-related fields) were retained and transformed into a consistent numeric representation; in particular, the image size information, originally encoded as a string representation of image width and height, was decomposed into two complementary numerical features capturing the total pixel area and the aspect ratio. Missing numerical values were handled via feature-wise mean imputation, and all numerical metadata were scaled to the [0,1] range using Min–Max normalization to prevent scale imbalance and support stable optimization.

In addition, selected categorical EXIF-related attributes (e.g., encoding and subsampling descriptors, as well as fingerprint-like fields) were encoded using one-hot encoding. Missing values in categorical metadata were explicitly encoded using a dedicated “missing” category, allowing the absence of categorical

EXIF information to be represented explicitly in the model input. The resulting numerical and one-hot encoded categorical vectors were concatenated to form a single metadata representation per image.

A hybrid multimodal classifier was constructed by combining a CNN-based visual feature extractor with a metadata processing module. Image classification is now formulated as a multimodal learning problem in which structured metadata are combined with visual representations through a late-fusion architecture. Visual features extracted by the CNN backbone are complemented by an independent metadata processing branch, implemented as a lightweight multi-layer perceptron (MLP). This metadata branch is designed to model interactions among heterogeneous EXIF attributes and to project them into a compact embedding space compatible with the dimensionality of the visual features.

The fusion between modalities is performed at the representation level by concatenating the metadata embedding with the CNN-derived visual embedding, forming a joint feature vector that is subsequently fed to a final linear classification head. This late-fusion strategy allows each modality to be processed independently while enabling the classifier to exploit cross-modal correlations at a higher semantic level. In most configurations, the visual backbone is kept frozen during training, concentrating learning capacity on the metadata MLP and the classification layers. This design encourages the model to leverage complementary EXIF information without overwhelming or distorting the visual representation, improving robustness in cases where pixel-level evidence alone is insufficient for reliable discrimination.

4.3.2 CNN Architecture with Social Media Metadata Integration

For this model, visual representations extracted by the CNN are augmented with contextual information derived from the social media environment in which images are shared. Social metadata are processed through modality-specific branches to obtain compact embeddings that capture complementary non-visual cues. Textual information from image captions is encoded using a pretrained multilingual BERT model. Captions are tokenized and truncated to a fixed maximum length, and the resulting token sequences are passed through the frozen BERT encoder. A single fixed-length text representation is obtained from the pooled embedding provided by BERT's pooler output, which summarizes the semantic content of the caption. This high-dimensional text embedding is then projected through a linear transformation to match the dimensionality of the visual feature space.

In parallel, numerical social signals such as engagement-related features are processed through a lightweight feed-forward network to produce a complementary metadata embedding. The projected text embedding, numerical metadata embedding, and CNN-derived visual representation are concatenated at the representation level to form a joint feature vector. This late-fusion architecture allows each modality to be processed independently while enabling the final classification head to exploit cross-modal correlations. During training,

learning is primarily concentrated on the projection, fusion, and classification layers, while the visual backbone and text encoder remain frozen. This design choice reduces overfitting and ensures that contextual social signals complement, rather than override, intrinsic visual information when discriminating between AI-generated and real images.

4.3.3 CNN Architecture with Social Media and EXIF Metadata Integration

The last model extends the multimodal framework by jointly integrating intrinsic image metadata and external social media context within a unified late-fusion architecture. Visual features extracted by the CNN backbone are complemented by two parallel metadata branches: one processing structured EXIF attributes through a lightweight MLP, and another encoding social media information, including caption text and numerical engagement signals. Textual captions are represented using the frozen pretrained BERT encoder and projected into a compact embedding space, while numerical social features are processed through a dedicated feed-forward network.

The embeddings derived from the visual, EXIF, and social metadata branches are concatenated at the representation level to form a joint feature vector, which is then passed to a final classification head. This design allows the model to exploit complementary signals from image content, intrinsic image level metadata, and social context while preserving modality-specific representations. By combining intrinsic and contextual information, the model aims to improve robustness and discrimination performance in challenging cases where either visual or metadata cues alone may be insufficient.

5 Results

In this section we show the accuracy, precision, recall, and F1 scores for our different classifiers. We first compare the out-of-the-box CNN model to our finetuned CNN version, both with and without the SSL loop, so that we can establish the effect of finetuning the CNN model and view the contribution of our SSL approach (see Table 1). Next, we investigate the effect of adding metadata to help detect AI-generated images. For this, we compare the finetuned CNN model with the SSL approach to the same model with the addition of EXIF metadata, social media metadata, and both types of metadata. An overview of the models' performances can be found in Table 2. The confusion matrices of every model can be found in Appendices A.3 & A.3.

Metric	Baseline CNN	Finetuned CNN	Finetuned CNN + SSL
Accuracy	0.500	0.720	**0.860**
Precision	0.439	0.650	**0.933**
Recall	**0.900**	0.650	0.700
F1 Score	0.590	0.650	**0.800**

Table 1: Comparison of classification metrics across CNN models.

Metric	CNN	CNN + EXIF	CNN + Social Media	CNN + EXIF + Social
Accuracy	0.860	0.860	0.796	**0.878**
Precision	**0.933**	0.760	0.708	0.769
Recall	0.700	0.950	0.850	**1.000**
F1 Score	0.800	0.844	0.772	**0.867**

Table 2: Classification metrics of the finetuned CNN with semi-supervised learning compared to metadata-augmented variants.

6 Discussion

As we can see in 1, there is a big performance decrease of the pretrained CNN on our test set compared to what was reported by the authors, which reported an over 99% accuracy on a perfectly class-balanced dataset. The class balance can be different on social media, and this is reflected by the results on our dataset. This showcases the difficult nature of the problem at hand; social media may contain a wide variety of data, making it difficult for CNN-based models to correctly classify content. These results showed us that we cannot directly use the out-of-the-box CNN model to attempt to label our unlabeled data, as it suggests that too many pseudo-labels would be unreliable, and could thus potentially negatively impact the rest of the training.

We can also observe the impact of finetuning as well as using a semi-supervised learning framework in Table 1. We see that not only finetuning but also an SSL approach leads to a great performance increase of the CNN model. This stems mainly from the size of the available dataset; without using SSL, the size of the dataset is so small that it is very difficult to extract meaningful patterns from it without overfitting such a small set. SSL enables the model to greatly increase the size of the available dataset while not sacrificing too much of the reliability of the dataset. To do this, we experimented with different decision thresholds to determine what was best. We found a threshold of around 0.98 to work best. A lower threshold led the dataset to grow too big too quickly (i.e., with too many inaccurate labels), and even higher thresholds meant the standard was so high that too few labels were accepted to significantly change the number of available instances. We also found this threshold to lead to a steady increase in dataset size, as opposed to a sharp increase in the beginning, which then plateaus. This implies that the labels of the unlabeled data are learned more gradually, potentially meaning a better learning process.

Table 2 next compares the performance of the finetuned CNN with different kinds of metadata incorporated. The finetuned CNN using semi-supervised learning can be considered the baseline of these models, which achieves high accuracy (0.860) and the highest overall precision (0.933), indicating that the incorporation of confidently pseudo-labeled data substantially improves the model's ability to correctly identify AI-generated images while minimizing false positives. However, this improvement in precision comes at the cost of a reduced recall (0.700), suggesting that the model becomes more conservative and fails to detect a substantial portion of AI-generated content. In contrast, augmenting the CNN with EXIF metadata leads to a notable increase in recall (0.950) at the expense of precision (0.760), indicating a stronger sensitivity to AI-generated images but with an increased rate of false positives. A similar trend is observed when incorporating social media metadata, although both precision and recall are slightly lower, resulting in the weakest overall F1 score among the metadata-augmented models. Combining both EXIF and social media metadata yields the best overall performance, achieving perfect recall (1.000) while also improving precision relative to both EXIF and social media metadata alone, although still not surpassing the precision achieved by the SSL-only CNN model. These re-

sults can be well illustrated using some example misclassifications from the test set, as seen in Figure 4 and Figure 5 in the Appendix. Here, we see that some realistic AI images are predicted as not AI despite being AI-generated, whereas in the metadata enhanced models some real somewhat unconventional pictures are wrongly labeled as AI. Overall, it appears as though all of the metadata-enhanced models tend to favor recall over precision and can outperform the CNN that does not have metadata included. This shows that metadata may be used to improve a CNN model to further improve performance.

6.1 Limitations

Overall, it can be seen that metadata can offer a different perspective and can improve a CNN model's classification performance. However, we believe that with such a small data set the results might not accurately reflect the real world performance. Here, we were only able to use the images for which metadata was available. This was not the case for many of the originally labeled images, which means these models heavily rely on pseudo-labeled data. We suspect that being able to use more labeled instances might help further improve performance, as well as get more realistic results. Furthermore, the component of the combined model that processes the metadata needs to be trained from scratch. This means that it requires substantial amounts of data until it can have a strong impact on the pipeline. We theorize that with our small dataset, we are not able to get the best possible performance out of it, and we suspect with more data all three models could be further improved.

Another limitation worth mentioning is that the data collected from Instagram was scraped from a single user's account. Since current social media platforms tailor the content to each individual user, we believe that this approach only exposes part of the additional challenges presented by content seen online. We believe that gathering content from numerous diverse users would make the pipeline more robust.

Moreover, images were gathered from different subfolders of the *Hashtag* directory, corresponding to hashtags such as *ai*. We believe these images are less likely to deceive users (and thus easier to classify) as AI, which means our results might be overly optimistic.

A final limitation has to do with the watermark removal process for our data pre-processing. We are unsure how to deal with impainted regions of an image and are aware the cropping out of logos has affected the image ratios of those images. This may also have impacted our results.

In summary, our results show great promise for both semi-supervised learning approaches as well as metadata incorporation. However these methods need to be tested rigorously on larger and more diverse datasets before being employed by social media platforms for AI content detection.

6.2 Future Work

A natural continuation of our work entails testing these approaches on larger and more diverse datasets. Even though the results here appear promising, the dataset used might fail to expose further challenges that need to be addressed before applying our methods at a large scale. A further possible continuation would be to analyze whether stacking of these models may provide a more robust AI detection pipeline. A possibility would be to run content through all of these classifiers separately and take the average prediction to determine whether an image is AI-generated or not.

6.3 Conclusion

Overall, we are able to conclude that semi-supervised learning can be used with great success to reliably pseudo-label instances and in turn improve the performance of a CNN to detect AI images in social media datasets. We see that with a well-chosen confidence threshold, the model is able to use the unlabeled data to identify more patterns in the data, ultimately leading to better performance. Furthermore, we see that metadata-augmented models can outperform a finetuned CNN model. Especially a combination of both EXIF and social media metadata proved to be the best performing classifier among all variants. Although our dataset is very small, these results seem promising. Further research should finetune these models on larger datasets to achieve an even better performance and explore other possible challenges with social media content classification. Thereafter these models may be employed by social media platforms for improved AI detection in the wild.

7 Team Contribution

The table below provides an overview of the contributions of each team member during the final phase of the project. All team members participated discussion and iterative refinement of the report.

Team Member	Task Description
Merel	Abstract, Introduction, Dataset, Methodology and Discussion sections. Contributed to dataset construction, labeling, image loading and preprocessing, watermark removal procedure, and overall report editing and consistency checks.
Michal	Methodology, Results, Discussion, Limitations, Further Research and Conclusion sections. Designed and implemented the semi-supervised learning (SSL) framework, CNN training pipeline, pseudo-labeling strategy, class weighting scheme and desgin and hyperparameter tuning of metadata-based models.
Ezgi	Introduction, Research Context, Dataset, and Appendix sections; Contributed to dataset construction, labeling, image loading and preprocessing, watermark removal procedure, and overall report editing.
Paola	Methodology, Research Context, Results and Appendix sections. Worked on metadata extraction and preprocessing (EXIF + Social Media), design and implementation of multimodal architectures, hyperparameter tuning and training of metadata-based models.
Kutay	Research Context and Methodology sections. Implemented CNN fine-tuning experiments.

Table 3: Task distribution among group members in the final project phase.

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A Appendix

A.1 Additional Data Cleaning Examples



(a) Original image with border watermark.

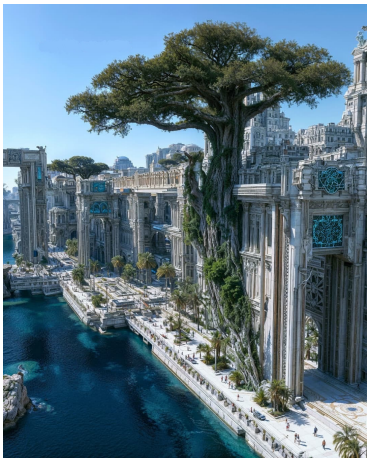


(b) Image after watermark removal via cropping.

Figure 2: Example of watermark removal via cropping when the watermark appears near image borders.



(a) Original image with centrally located watermark.



(b) Image after watermark removal via inpainting.

Figure 3: Example of watermark removal via masking/inpainting when cropping would remove important visual content.

A.2 SSL methodology pseudocode

Algorithm 1 Semi-Supervised Learning loop

```
Split labeled data into training and validation sets
for i = 1 to n_iterations do
    Add all pseudolabeled images to the training set
    Finetune model on training set
    Evaluate train set metrics
    Evaluate validation set metrics
    for each unlabeled instance do
        Perform inference using the CNN model
        if confidence > threshold then
            Assign model prediction as pseudolabel
        end if
    end for
end for
Compute metrics on test set
```

Algorithm 2 Finetuning loop

```
Initialize best validation loss = ∞
Initialize best model state = None
Initialize epochs without improvement = 0
for i = 1 to n_epochs do
    Train model on training data
    Compute training loss and update optimizer
    Perform inference on validation set
    Compute validation loss
    if validation loss < best validation loss then
        best validation loss = validation loss
        best model state = current model state
        epochs without improvement = 0
    else
        epochs without improvement += 1
    end if
    if epochs without improvement > patience then
        break
    end if
end for
return best model state
```

A.3 Confusion matrices of the models

True / Pred	AI	Human
AI	18	2
Human	23	7

Table 4: Confusion matrix for the out-of-the-box CNN baseline model.

True / Pred	AI	Human
AI	15	5
Human	8	22

Table 5: Confusion matrix for the finetuned CNN without semi-supervised learning.

True / Pred	AI	Human
AI	14	6
Human	1	29

Table 6: Confusion matrix for the finetuned CNN with semi-supervised learning.

True / Pred	AI	Human
AI	19	1
Human	6	24

Table 7: Confusion matrix of the CNN model with EXIF metadata.

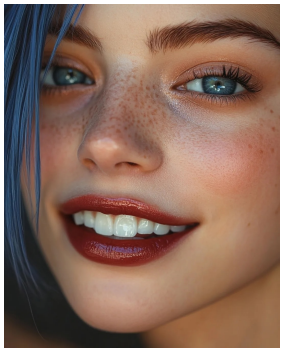
True / Pred	AI	Human
AI	17	3
Human	7	22

Table 8: Confusion matrix of the CNN model with social media metadata.

True / Pred	AI	Human
AI	20	0
Human	6	23

Table 9: Confusion matrix of the CNN model with EXIF + social media metadata.

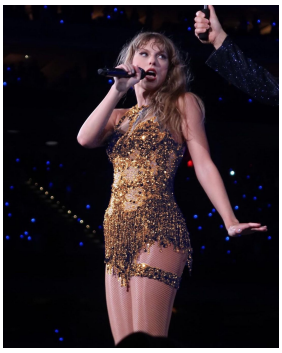
A.4 Visualization of Misclassified Samples Across All Models



(a) Finetuned CNN: True Label: AI, Predicted Label: Not AI



(b) Finetuned CNN: True Label: AI, Predicted Label: Not AI



(c) CNN + EXIF: True Label: Not AI, Predicted Label: AI

Figure 4: Misclassified Images



(a) CNN + EXIF: True Label: Not AI, Predicted Label: AI



(b) CNN + Social Media: True Label: AI, Predicted Label: Not AI



(c) CNN + EXIF + Social Media: True Label: Not AI, Predicted Label: AI

Figure 5: Misclassified Images